

1)What is the main advantage of requiring a smaller number of bits to represent the range of unique sequence numbers employed by a protocol?

Daha az sayıda bit gerektirmenin en önemli avantajı nedir,Bir protokol tarafından kullanılan benzersiz sıra numaralarının aralığını göstermek için

Less overhead/header in the packet:daha az maliyet/paket başlığı içersindeki

more efficient use of the bandwidth:Bant genişliğinin daha verimli kullanılması

2)write down the names of five mechanisms for correct data transfer?

Doğru veri aktarımı için beş mekanizmanın adlarını yazın

Time-out

Retransmission

Error Control(checksum)

Acknowledgment

Sequence Number

3)what is the window size in Stop and Wait Protocol? Stop and Wait protokolünün pencere boyutu nedir?

Window Size=1 only one packet can be sent at a time

pencere boyutu=1 sadece tek bir paket gönderilebilir aynı anda

4)How many unique sequence numbers does Stop and Wait need?

Stop and wait protokolü kaç tane sıra numarasına ihtiyaç duyar

1 bit-->2 sequence numbers

5)Vize sorusu:Stop and Wait no need sequential number?

False-->Stop and Wait needs 1 bit-->2 sequence numbers

6)Vize sorusu TCP is always better UDP except header application...?

FALSE:UDP is faster than TCP(UDP TCP den daha hızlıdır)

7)Vize sorusu :UDP error mechanism can detect all error but not correct them?

UDP hata mekanizması bütün hataları belirleyebilir fakat doğrulayamaz?

FALSE:UDP Error mechanism is Checksum.can not detect 2 bits error,if they are in same column or different rows

UDP hata mekanizması checksum'dır.2 bitlik hataları belirleyemez,aynı sütün veya farklı satırda olanları.

6)Vize sorusu:Go-Back N out of order the packets? Go back N paketleri sıra dışı eder mi?

TRUE:Go Back N is cumulative,out of order packets will be discard and will not buffer

TRUE:Go Back N cumulative dir,sırasız gelen(out of order) paketleri göz ardı eder(discard) ve depolanmaz(no buffer)

9)Write Down the names of six fields in TCP Segment Structure?

TCP Segment yapısındaki 6 alanı yazınız?

Destination Port-Source Port-ACKNOWLEDGE Number-Sequence Number-RST-ACK

10)Write down four properties of TCP? TCP'nin dört özelliğini yazın?

Connection Oriented:bağlantı yönelimli

Point to point connection:noktadan noktaya bağlantı

Full duplex service

3 way handshaking

does not support multicasting:Çok noktaya yaymayı desteklemez

11)FALSE:If the PSH bit in TCP segment is set,the receiver should pass the data to the network layer immadiately?

The receiver should pass to the data to Application Layer

Network Layer değil Application Layer olacak

bazen soruda sender verir,receiver olmalı.

yani doğru olan **receiver** ve **application layer** birlikte olacak

12)With Go-Back-N,it is possible for the sender to receive an ACK for a packet that falls outside of its current window?

Go-Back-N ile göndericinin geçerli penceresinin dışına çıkan bir paket için bir ACK alması mümkün müdür?

TRUE:Go Back N is using cumulative ACK

p1,p2,p3 diye 3 paketim var P1 ve p3 karşı tarafa ulaştı p2 ulaşamadı,dolayısıyla

ACK 1 i iki defa gönderecem p2 nin time out süresi içinde(şekil için)

1)Write down 5 differences between TCP and UDP?

TCP:Slower-flow control-congestion control-guarantee for arrival of data-3 way handshake-20 byte header-

UDP:Faster-No Flow Control-no congestion Control-No guarantee for arrival of data-no 3 way handshake-8byte header

2)What is the difference between GO Back N and Selective Repeat?

Go Back N:Receiver does not have a buffer

Selective Repeat:receiver has a buffer

3)UDP has a checksum mechanism,but UDP does not provide correct data transfer?why not

Checksum is used for error control,but it can not find error point and can not edit

Farklı cevap:Since UDP does not use 3 way handshake,It does not guarantee reliable data transfer

4)What is the disadvantage of a Stop and Wait protocol?what is the solution to the problem?

Only one packet can be sent at a time,use pipelined protocols(can send multiple packets at a time)

5)What is the importance of a congestion control algorithm?

because Congestion control mechanism prevent packet loss

Congestion avoidance,fast recovery,fast retransmit,no overflow

6)in TCP there is a Slow Start phase,why is called'slow start'?

because it starts from 1,minimum is 1 in TCP congestion control phase

7)What is the difference between TCP Tahoe and TCP Reno?

TCP Reno has not slow start phase,but Tahoe has slow start phase.

8)Explain the meanings of flag bits in the TCP segment?

URG=urgent field is significant

PSH=push function

ACK=indicates that ACK field significant

RST=reset the connection

SYN=synchronize sequence numbers

FIN=no more data from sender

9)Why is flow control important?

because it prevents packet loss,no overflow

10)Sequence number=is the byte stream number of the first byte ,nt he segment

ACK Number=is the sequence number of the next byte the receiver is expecting from sender.